

Year 12 Mathematics Term 2 Assessment 2004

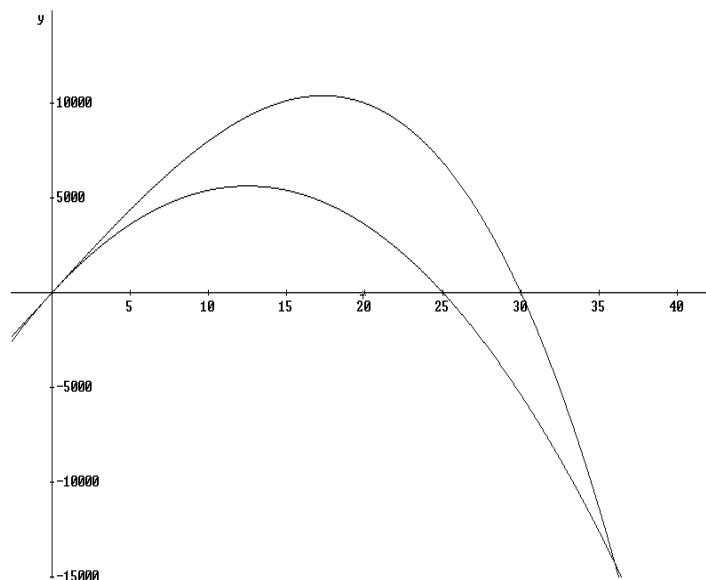
QUESTION 1

- (a) The population, P , of a town at time t years after the start of 1970 is estimated by the formula $P = P_0 e^{kt}$. Records show that the population of the town at the start of 1970 was 2916 and it had grown to 4860 by the start of 1980.
- (i) Find the exact values for P_0 and k . 3
- (ii) Find the population of the town at the start of 2010. (Give answer correct to nearest 100 people) 2
- (iii) Prove that the rate of change of the population at time t years is given by $\frac{dP}{dt} = kP$ where k is a constant. 1
- (iv) Find the rate of increase of the population
- (α) when the population is 6000. (Give answer correct to 2 significant figures) 1
- (β) at the beginning of the year 2000. (Give answer correct to 2 significant figures) 1

- (b) The graph shows the positions (x km) of two objects A and B at time t hours. The expressions for the positions are given by:

$$x_A = 900t - t^3$$

$$x_B = 900t - 36t^2$$



- (i) Find the time interval in which the position of each object is to the right of the origin. 2
- (ii) Find an expression (in terms of t) for the distance (D) between the two objects while they are both to the right of the origin 1
- (iii) Find the greatest distance between the two objects while they are to the right of the origin 4

QUESTION 2**(START A NEW PAGE)**

(a) Sally is given 3 Mars Bars. With each Mars Bar there is a 20% chance of winning a free Mars Bar.

(i) Draw a probability tree diagram for the above information.

1

What is the probability that Sally wins

(ii) no free Mars Bars?

1

(iii) at least 1 free Mars Bar?

1

(iv) exactly 1 free Mars Bar?

1

(b) (i) Sketch the parabola $y = 2x^2 + 5x - 12$ clearly showing all intercepts with the coordinate axes.

2

(ii) Hence or otherwise solve for p such that $2p^2 + 5p > 12$.

1

(c) An object initially at the origin moves with velocity ($v \text{ km/h}$) given by $v = 24 + 10t - t^2$. Find

(i) the maximum speed of the object.

2

(ii) the acceleration when the object is at rest.

3

(iii) the total distance travelled by the object during the first 15 hours.

3

QUESTION 4**(START A NEW PAGE)**

(a) The roots of the equation $3x^2 - 5x + 1 = 0$ are $x = \alpha$ and $x = \beta$. Find the value of

(i) $\alpha + \beta$. 1

(ii) $\alpha\beta$. 1

(iii) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$. 2

(b) On a table are two jars. The red jar contains 5 cards numbered 0, 2, 4, 6 and 8 while the blue jar contains 5 cards numbered 1, 3, 5, 7 and 9. A card is drawn from each jar and the sum of the two cards is calculated.

(i) Draw a dot diagram illustrating the above information. 1

Find the probability that

(ii) the sum is more than 10. 1

(iii) the sum is a prime number. 1

(iv) the sum is more than 10 if it is known that the sum is prime. 2

(c) The acceleration ($\ddot{x} \text{ ms}^{-2}$) of a particle at time t seconds is given by $\ddot{x} = \frac{-1}{(5t+1)^2}$. The particle is initially 3 metres to the left of the origin traveling with velocity 1 ms^{-1} .

(i) Find an expression for the particle's velocity (v) at time t . 2

(ii) Find the limiting speed of the particle. 1

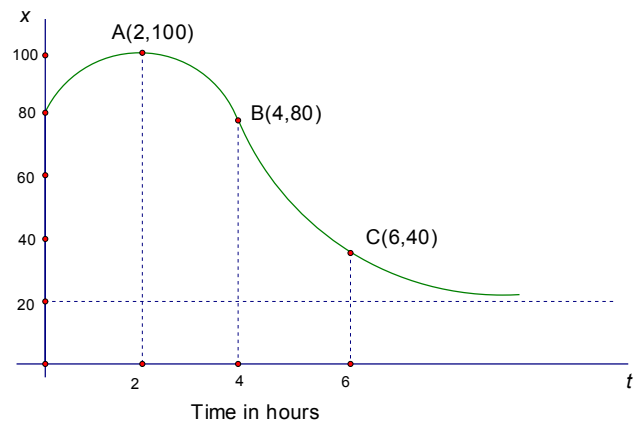
(iii) Determine the position of the particle at the end of the tenth second. (Give answer correct to the nearest metre) 3

QUESTION 3

(START A NEW PAGE)

- (a) The displacement-time graph for an object is shown.

The graph has a turning point at $A(2,100)$ and an inflexion point at $B(4,80)$.



- (i) Find when the object changes direction. 1
 - (ii) Find the position of the object when its speed is greatest. 1
 - (iii) Find the time interval for which the acceleration is negative. 2
 - (iv) Briefly describe the motion of the object between the points A and B . 2
- (b) A scientist studying a penguin colony estimates that the number $N(t)$ of penguins in the colony at the end of t years is given by the formula $N(t) = \frac{A}{5 + 3e^{-0.1t}}$ where A is a constant.
- (i) When the scientist starts her study the penguin population is estimated at 12 000. Find the value of A . 1
 - (ii) Find the estimated population of the colony at the end of 8 years. (Give your answer correct to the nearest 100 penguins) 1
 - (iii) Find the time required for the colony to grow to 18 000 penguins. (Give your answer correct to 1 decimal place) 2
 - (iv) Approximately how many penguins would you expect to find in the colony after a long time? 1
- (c) (i) If the line $y = mx + b$ is tangent to the hyperbola $y = \frac{a}{x}$ prove that $b^2 + 4am = 0$. 2
- (ii) Hence find the values of b for which the line $y = b - 3x$ will always intersect with the hyperbola $y = \frac{12}{x}$. 2

Term 2 2u 2004

Growth/Decay

Question 1

The population (P) of a town at time t years after the start of 1970 is estimated by the formula $P = P_0 e^{kt}$. Records show that the population of the town at the start of 1970 was 2916 and it had grown to 4860 by the start of 1980.

- (i) Find the exact values for P_0 and k .
- (ii) Find the population of the town at the start of 2010.
- (iii) Prove that the rate of change of the population at time t years is given by $\frac{dP}{dt} = kP$ where k is a constant.
- (iv) Find the rate of increase of the population
 - (α) at the beginning of the year 2000.
 - (β) when the population is 6000.

Question 2

A scientist studying a penguin colony estimates that the number $N(t)$ of penguins in the colony at the end of t years is given by the formula $N(t) = \frac{A}{5 + 3e^{-0.1t}}$ where A is a constant.

- (i) When the scientist starts her study the penguin population is estimated at 12 000. Find the value of A .
- (ii) Find the estimated population of the colony at the end of 8 years. (Give your answer correct to the nearest 100 penguins)
- (iii) Find the time required for the colony to grow to 18 000 penguins.
- (iv) Approximately how many penguins would you expect to find in the colony after a long time?

Probability

Question 1

George has forgotten his 5 digit security number. He remembers that it is an odd number, no digits are repeated and it has alternating odd and even numbers.

- (i) How many security numbers satisfy the above conditions?
- (ii) Find the probability that his security number is greater than 85 000.

Question 2

Mary is given 3 Mars Bars. With each Mars Bar there is a 20% chance of winning a free Mars Bar.

- (i) Draw a probability tree diagram for the above information.

What is the probability that Mary wins

- (ii) no free Mars Bars?
- (iii) at least 1 free Mars Bar?
- (iv) exactly 1 free Mars Bar?

Question 3

On a table are two jars. The red jar contains 5 cards numbered 0, 2, 4, 6 and 8 while the blue jar contains 5 cards numbered 1, 3, 5, 7 and 9. A card is drawn from each jar and the sum of the two cards is calculated.

- (i) Draw a dot/lattice diagram illustrating the above information.

Find the probability that

- (ii) the sum is more than 10?
- (iii) the sum is a prime number?
- (iv) the sum is more than 10 if it is known that the sum is prime?

Motion

Question 1

An object initially at the origin moves with velocity ($v \text{ km/h}$) given by $v = 24 + 10t - t^2$. Find

- (i) the initial velocity of the object.
- (ii) the maximum speed of the object.
- (iii) the acceleration when the object is at rest.
- (iv) the distance traveled by the object during the first 15 hours.

Question 2

The position (x m) of a particle at time t seconds is given by $x = 100t + 750e^{-0.05t}$.

- (i) Find expressions for the velocity (v) and acceleration (a) at time t .
- (ii) Find the initial position and velocity of the particle.
- (iii) Find the time taken for the particle to reach a speed of 180 m/s.
- (iv) Prove that $a = 0.05(100 - v)$.

Question 3

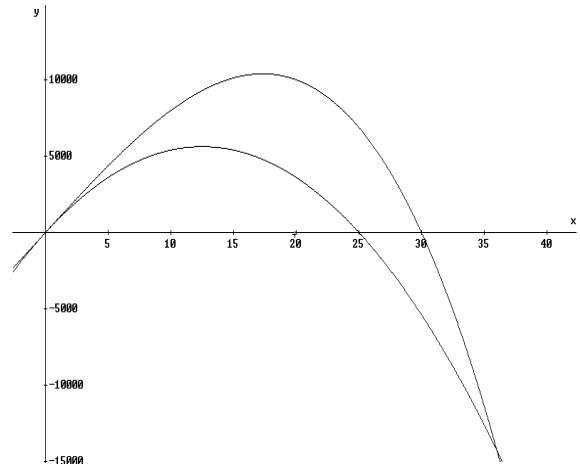
The position of an object at time t is given by $x = 3 + 4\sin^2 t$.

- (i) Find expressions for its velocity and acceleration at any time t .
- (ii) Prove that $\ddot{x} = 4(5 - x)$.
- (iii) Find the smallest value of t for which the acceleration is zero.

Question 4

The graph shows the positions of two objects A and B at time t are given by

$$x_A = 900t - t^3$$
$$x_B = 900t - 36t^2$$



- (i) Find the time interval in which the position of each object is to the right of the origin.
- (ii) Find an expression (in terms of t) for the distance (D) between the two objects while they are both to the right of the origin
- (iii) Find the greatest distance between the two objects while they are to the right of the origin.

Quadratics

Question 1

Find the roots of the equation $2y^2 - 6y + 3 = 0$. Give your answer in simplest form.

Question 2

- (i) Sketch the parabola $y = 2x^2 + 5x - 12$ clearly showing all intercepts with the coordinate axes.
- (ii) Hence or otherwise solve $2p^2 + 5p - 12 > 0$.

Question 3

The roots of the equation $3x^2 - 5x + 1 = 0$ are $x = \alpha$ and $x = \beta$. Find the value of

- (i) $\alpha + \beta$.
- (ii) $\alpha\beta$.
- (iii) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} + 2$.

Question 4

(i) If the line $y = mx + b$ is tangent to the hyperbola $y = \frac{a}{x}$ prove that $b^2 + 4am = 0$.

(ii) Hence find the values of b for which the line $y = b - 3x$ always intersects the hyperbola $y = \frac{12}{x}$.

Question 5

(i) Prove that $(m - n)^2 = (m + n)^2 - 4mn$.

(ii) Given that $x = \alpha$ and $x = \beta$ are roots of the quadratic equation $px^2 + qx + 1 = 0$ and $\alpha > \beta$, find an expression for $\alpha - \beta$. Express your answer in simplest form.

Question 6

Find the value(s) of k for which ***** is positive definite

Question 7

Show that ***** always has rational roots if ***** are rational.

Question 8

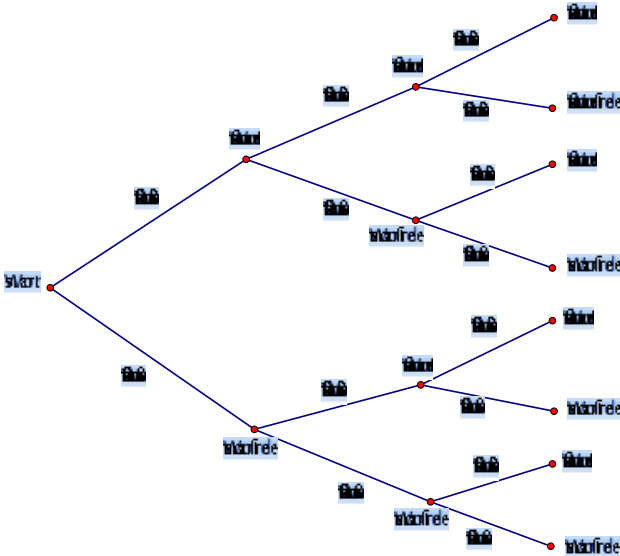
Find the minimum value of y if $y = 3x^2 - 8x + 6$.

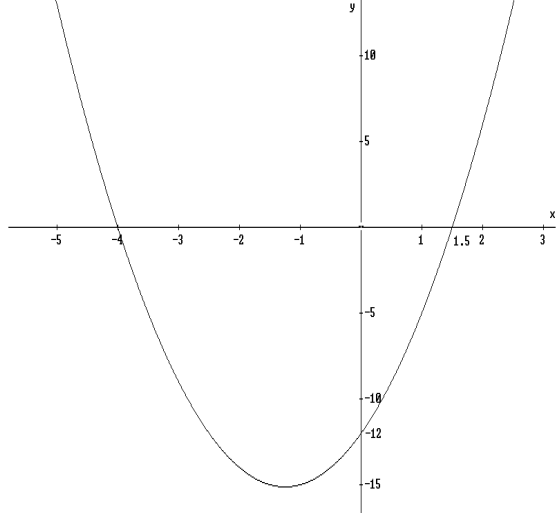
Year 12 Term 2 Mathematics Assessment 2004 - Solutions

Question 1			
(a)	(i)	when $t = 0$, $2916 = P_0 e^0$ $\therefore P_0 = 2916$ when $t = 10$, $4860 = 2916 e^{10k}$ $e^{10k} = \frac{5}{3}$ $10k = \ln\left(\frac{5}{3}\right)$ $k = \frac{1}{10} \ln\left(\frac{5}{3}\right)$	3
	(ii)	when $t = 40$, $P = 2916 e^{4 \ln\left(\frac{5}{3}\right)}$ $\therefore$ population = 22 500	2
	(iii)	$\frac{dP}{dt} = k P_0 e^{kt}$ $= kP$ since $P = P_0 e^{kt}$	1
	(iv)	(α) when $P = 600$, $\frac{dP}{dt} = \frac{1}{10} \ln\left(\frac{5}{3}\right) \times 600$ $= 310$ (to 2 sig. fig.)	1
		(β) when $t = 30$, $P = k P_0 e^{kt}$ $= \frac{1}{10} \ln\left(\frac{5}{3}\right) \times 2916 \times e^{3 \ln\left(\frac{5}{3}\right)}$ $= 690$ (to 2 sig. fig.)	1
(b)	(i)	when $x_A = 0$, $t(30 - t)(30 + t) = 0$ $t_A = 0$ or ± 30 when $x_B = 0$, $36t(25 - t) = 0$ $t_B = 0$ or 25 Time intervals are $0 < t_A < 30$ and $0 < t_B < 25$.	2
	(ii)	$D = x_A - x_B$ $= (900t - t^3) - (900t - 36t^2)$ $D = 36t^2 - t^3$	1

	(iii)	$\frac{dD}{dt} = 72t - 3t^2$ for stat. pt. $\frac{dD}{dt} = 0$ $72t - 3t^2 = 0$ $3t(24 - t) = 0$ $t = 0 \text{ or } 24$ $\frac{d^2D}{dt^2} = 72 - 6t$ when $t = 0$, $\frac{d^2D}{dt^2} = 72 > 0$ $\therefore$ local min. tp. when $t = 24$, $\frac{d^2D}{dt^2} = -72 < 0$ $\therefore$ local max. tp. when $t = 24$, $D = 36(24)^2 - (24)^3$ $= 6912$ maximum distance is 6912 m.	4
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Question 2

(a)	(i)		1
	(ii)	$P(\text{no free Mars bar}) = (0.8)^3 \text{ or } \left(\frac{4}{5}\right)^3$ $= 0.512 \text{ or } \frac{64}{125}$	1
	(iii)	$P(\text{at least one free Mars bar}) = 1 - P(\text{no free Mars bars})$ $= 1 - (0.8)^3$ $= 0.488 \text{ or } \frac{61}{125}$	1
	(iv)	$P(\text{one free Mars bar}) = 3 \times (0.2) \times (0.8)^2 \text{ or } 3 \times \left(\frac{1}{5}\right) \times \left(\frac{4}{5}\right)^2$ $= 0.384 \text{ or } \frac{48}{125}$	1

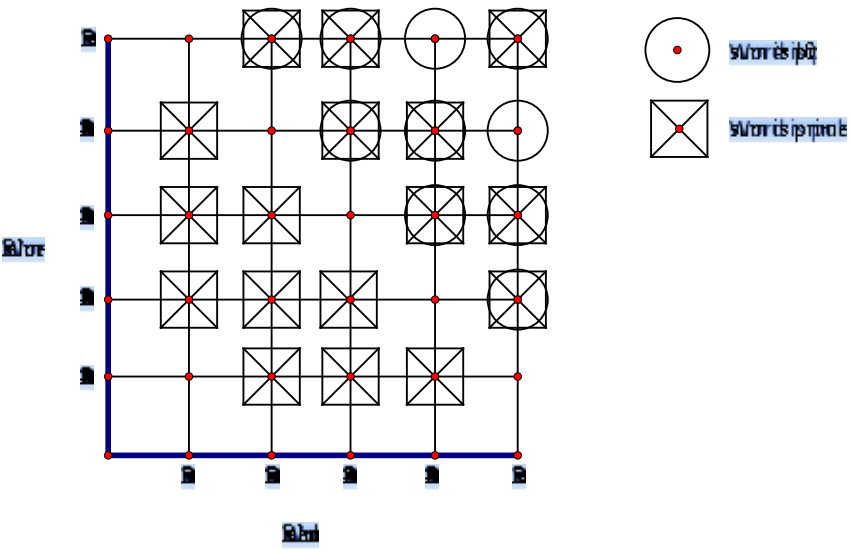
(b)	(i)	if $x = 0, y = -12$ if $y = 0, 2x^2 + 5x - 12 = 0$ $(2x - 3)(x + 4) = 0$ $x = \frac{3}{2}$ or -4		2
	(ii)	$\{p : p < -4\} \cup \{p : p > 1.5\}$ or $p < -4$ or $p > 1.5$		1
(c)	(i)	Since the expression for velocity is a quadratic with negative leading coefficient, the maximum speed is at the turning point $\therefore t = -\frac{10}{-2}$ $= 5$ $v = 24 + 10(5) - (5)^2$ $= 49$ $\therefore \text{max speed} = 49 \text{ km/hr}$		2
	(ii)	at rest when $v = 0$ $-t^2 + 10t + 24 = 0$ $t^2 - 10t - 24 = 0$ $(t - 12)(t + 2) = 0$ $t = 12 \quad (t \geq 0)$ $a = 10 - 2t$ when $t = 12, a = 10 - 24$ $= -14$ $\therefore \text{acceleration is } -14 \text{ km/hr}^2$		3

	(iii)	$x = 24t + 5t^2 - \frac{1}{3}t^3 + c$ when $t = 0$, $x = 0$ $0 = 0 + 0 + 0 + c$ $\therefore c = 0$ $x = 24t + 5t^2 - \frac{1}{3}t^3$ when $t = 0$, $x = 0$ when $t = 12$, $x = 24(12) + 5(12)^2 - \frac{1}{3}(12)^3$ $= 432$ when $t = 15$, $x = 24(15) + 5(15)^2 - \frac{1}{3}(15)^3$ $= 360$ distance travelled $= 432 + (432 - 360) \text{ km}$ $= 504 \text{ km}$	3
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Question 3

(a)	(i)	At 2 hours	1
	(ii)	At 80m to the right of the origin or at position B	1
	(iii)	$0 \leq t < 4$	2
	(iv)	The object is moving to the left with increasing speed.	2
(b)	(i)	when $t = 0$, $N = 12\,000$ $12\,000 = \frac{A}{5 + 3e^0}$ $A = 96\,000$	1
	(ii)	when $t = 8$ $N = \frac{96000}{5 + 3e^{-0.1 \times 8}}$ ≈ 15122.9 $= 15100 \text{ (to nearest 100)}$	1
	(iii)	when $N = 18000$ $18000 = \frac{96000}{5 + 3e^{-0.1t}}$ $5 + 3e^{-0.1t} = \frac{16}{3}$ $3e^{-0.1t} = \frac{1}{3}$ $e^{-0.1t} = \frac{1}{9}$ $-0.01t = \ln\left(\frac{1}{9}\right)$ $t = \frac{\ln\left(\frac{1}{9}\right)}{-0.1}$ $= 21.97$ time = 22.0 years (to 1 d.p.)	2

	(iv)	as $t \rightarrow \infty$, $e^{-kt} \rightarrow 0$ $\therefore N \rightarrow \frac{96000}{5}$ $N \rightarrow 19\,200$ population tends to 19200 penguins.	1
(c)	(i)	Curves meet when $mx + b = \frac{a}{x}$ $mx^2 + bx = a$ $mx^2 + bx - a = 0$ Now for the line to be a tangent, this equation has must have only one solution i.e. the $\Delta = 0$ $\therefore \Delta = b^2 - 4m(-a)$ $b^2 - 4m(-a) = 0$ $b^2 + 4ma = 0$	2
	(ii)	Let $m = -3$ and $a = 12$ $\therefore b^2 + 4(12)(-3) \geq 0$ $b^2 - 144 \geq 0$ $(b - 12)(b + 12) \geq 0$ $b \leq -12 \text{ or } b \geq 12$	2
Question 4			
(a)	(i)	$\alpha + \beta = \frac{5}{3}$	1
	(ii)	$\alpha\beta = \frac{1}{3}$	1
	(iii)	$\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta}$ $= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta}$ $= \frac{\left(\frac{5}{3}\right)^2 - 2\left(\frac{1}{3}\right)}{\left(\frac{1}{3}\right)}$ $= \frac{19}{3}$	2

(b)	(i)		1
	(ii)	$P(\text{sum} > 10) = \frac{10}{25}$ $= \frac{2}{5}$	1
	(iii)	$P(\text{sum is prime}) = \frac{17}{25}$	1
	(iv)	$P(\text{sum} > 10 \text{ given the sum is prime}) = \frac{8}{17}$	2
(c)	(i)	$\ddot{x} = -(5t+1)^{-2}$ $v = \frac{-(5t+1)^{-1}}{-5} + c_1$ $= \frac{0.2}{5t+1} + c_1$ when $t = 0, v = 1$ $\therefore 1 = 0.2 + c_1$ $c_1 = 0.8$ $v = \frac{0.2}{5t+1} + 0.8$	2
	(ii)	as $t \rightarrow \infty$ $v \rightarrow 0 + 0.8$ limiting speed is 0.8 m/s	1

	(iii) $x = \frac{0.2 \ln(5t+1)}{5} + 0.8t + c_2$ $= 0.04 \ln(5t+1) + 0.8t + c_2$ when $t = 0, x = -3$ $-3 = 0.04 \ln(1) + 0 + c_2$ $c_2 = -3$ $x = 0.04 \ln(5t+1) + 0.8t - 3$ when $t = 10$ $x = 0.04 \ln(51) + 8 - 3$ ≈ 5.157 position is 5m to the right of the origin.	3
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